

Asadullah Bin Rahman

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Research Interests: Image Processing, Machine Learning, Quantum Computing

Employment

- 07/25 – Present ♦ **Lecturer, Computer Science and Engineering**
World University of Bangladesh (WUB), Dhaka, Bangladesh
Roles: Undergraduate teaching, thesis & project supervision, competitive programming club mentoring, student counseling
- 07/23 – 07/25 ♦ **Research Assistant**
IoThink Lab, Department of Computer Science and Engineering, HSTU
Focus: Wavelet-based denoising and deep learning for medical image analysis; interdisciplinary dataset collection
Supervisor: Dr. Md. Abdulla Al Mamun 🔗

Education

- 07/23 – 07/25 ♦ **MSc in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Image Denoising using Wavelet Transform and Deep Learning Architectures
CGPA: 3.547/4.00 🔗
- 01/17 – 12/22 ♦ **BSc (Engineering) in Computer Science and Engineering**, Hajee Mohammad Danesh Science and Technology University (HSTU), Bangladesh
Thesis: MRI Brain Tumor Classification from Contrast-Enhanced MRI using Deep Learning
CGPA: 3.31/4.00 🔗

Skills

- Quantum Computing ♦ Proficient in Qiskit, Cirq, PennyLane, Classiq; applied experience in Quantum Algorithms, Quantum Machine Learning, Quantum Key Distribution
- Machine Learning ♦ Proficient in NumPy, Pandas, SciPy, Matplotlib, OpenCV, PyTorch, TensorFlow; applied experience in Image Processing, Computer Vision
- Programming ♦ C++, Java, Python; Data Structures, Algorithms, Databases; Solved 300+ problems (CodeForces, HackerRank, UVa)
- Languages ♦ Bangla (Native), English (IELTS Academic: **6.5**, CEFR Level: B2) 🔗
- Others ♦ Linux, Git, LaTeX

Research Publications

Conference Proceedings

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, “Mitigating Noise from Biomedical Images Using Wavelet Transform Techniques,” in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025, pp. 1–6. 🔗 DOI: [10.1109/ECCE64574.2025.11013062](https://doi.org/10.1109/ECCE64574.2025.11013062).
2. **A. B. Rahman**, M. Touhid Islam, M. R. Islam, M. Sohrawordi, and M. N. Sultan, “Enhanced Brain Tumor Classification from MRI Images Using Deep Learning Model,” in *2023 26th International Conference on Computer and Information Technology (ICCIT)*, 2023, pp. 1–6. 🔗 DOI: [10.1109/ICCIT60459.2023.10441064](https://doi.org/10.1109/ICCIT60459.2023.10441064).

Preprints

1. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Deep Learning Architectures for Medical Image Denoising: A Comparative Study of CNN-DAE, CADTra, and DCMIEDNet*, 2025. arXiv: [2508.17223](https://arxiv.org/abs/2508.17223) [eess.IV].  URL: <https://arxiv.org/abs/2508.17223>.
2. **A. B. Rahman**, M. Ibn Afjal, and M. A. Al Mamun, *Systematic Evaluation of Wavelet-Based Denoising for MRI Brain Images: Optimal Configurations and Performance Benchmarks*, 2025. arXiv: [2508.15011](https://arxiv.org/abs/2508.15011) [eess.IV].  URL: <https://arxiv.org/abs/2508.15011>.

Teaching

- Fall 2026
 - ◇ Object Oriented Programming (Batch 71C, 71D, 71E, Semester 4). CSE 06132120.
- Summer 2026
 - ◇ Compiler Design (Batch 63A, Semester 11). CSE 06134156.
 - ◇ Object Oriented Programming (Batch 70E, Semester 4). CSE 06132120.
 - ◇ Introduction to Programming (Batch 73B, Semester 1). CSE 06131209.
- Spring 2026
 - ◇ Object Oriented Programming (Batch 69A, Semester 4). CSE 06132120.
 - ◇ Introduction to Programming (Batch 70B, 71C, Semester 2). CSE 06131209.
- Fall 2025
 - ◇ Computer Fundamentals (Batch 71G, Semester 1). CSE 06111101.
 - ◇ Object Oriented Programming (Batch 68A, Semester 4). CSE 06132120.
 - ◇ Computer and Cyber Security (Batch 63B, Semester 9). CSE 06124158.

Thesis Supervision

- Summer 2026
 - ◇ **Explainable Deep Learning for Plant Disease Classification Using Convolutional Neural Networks @ WUB**: Comparison of CNN architectures for leaf disease classification on PlantVillage crops, with Grad-CAM saliency exposing the leaf regions behind each prediction.
 - ◇ **ClassSync Pro: IoT-Based Smart Classroom Automation System @ WUB**: LMS-scheduled control of classroom air-conditioning, fans, lighting and projectors with sensor-based occupancy validation; projected 30–40% cut in campus energy use.

Research and Projects

- 02/26
 - ◇ **Multilayered Security** : Implemented and simulated a research project on Multi-Layered Security System: Integrating Quantum Key Distribution with Classical Cryptography to Enhance Steganographic Security.
- 11/24
 - ◇ **Guava Fruit Disease Dataset @ IoThink Lab** : Collaborated on dataset collection for interdisciplinary research.
- 08/24
 - ◇ **Quantum Machine Learning Projects @ Womanium Program**: Four QML projects completed for *QML for Conspicuity Detection in Production*.
 - **QML for Anomaly Detection** : hybrid quantum-classical model for anomaly detection.
 - **Quantumvolutional Neural Networks** : hybrid quantum convolutional model for MNIST digit classification.
 - **Quantum Variational Classifier** : quantum classifier for penguin species classification.
 - **Quantum Regression Model** : learned and predicted the sine function on the interval $[0, 2\pi]$.

Professional Development

- 09/24 – 12/24
 - ◇ **QClass24/25 Fall Semester** : Achieved 96% in 3 ECTS graduate-level program on Quantum Algorithms and QKD.

Professional Development (continued)

- o6/24 – o8/24
 - ◇ **Womanium Quantum + AI 2024** : Coursework on Quantum Computing and AI.
 - o8/24
 - ◇ **QGSS 2024 – Quantum Excellence**  : Intensive quantum computing boot camp by IBM Qiskit.
 - ◇ **Classiq Diploma** : Advanced Quantum Algorithm design.
 - ◇ **PennyLane Diploma** : Quantum Machine Learning Challenge completion.
 - o7/24
 - ◇ **QNickel Diploma** : Quantum Algorithms workshop.
 - ◇ **QBronze Diploma** : Introductory Quantum Computing and Programming.
 - o9/20
 - ◇ **Python for Everybody Specialization** : Coursera specialization on Python fundamentals and data structures.

Awards and Achievements

- o6/25
 - ◇ **Unitary Hack 2025**  : For open source contributions to the quantum software ecosystem.
- o4/25
 - ◇ **YQuantum 2025**  : Solved BlueQubit's Peaked Circuits Challenge.
- o2/25
 - ◇ **MIT iQuHACK 2025** : Solved IonQ's Max Cut problem.
- o8/24
 - ◇ **Womanium Quantum + AI 2024** : Program finalist and QSL fellowship nominee.
- o6/24
 - ◇ **IBM Quantum Challenge 2024**  : Ranked 11th globally.

Volunteering Work

- o4/25 – Present
 - ◇ **Coordinator & Mentor, QBangladesh (QWorld)** : Mentoring students and promoting quantum computing literacy in Bangladesh.
Hosted Events: QBronze , QSilver .

Hobbies

- ◇ Chess 
- ◇ Gardening